

Running in the Heat: Performance Consequences and Strategies to Prepare for Hot-Weather Racing

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ABSTRACT

Competitive runners seek to control as many performance variables as possible when preparing for peak competition. However, race day weather is out of an athlete's control and can have a profound impact on performance. This review defines hot weather in terms of running performance, details the physiology underlying heat-related performance decrements, and discusses strategies that may mitigate the negative impact of heat on cardiovascular strain and attenuate reductions in heat-related running performance.

WHAT IS HOT-WEATHER RUNNING?

Classic laboratory-based studies often subject athletes to environmental temperatures between 35 and 45°C with varying humidity to examine the impact of heat stress on physiological responses and exercise performance. The wet bulb globe temperature (WBGT; an index that accounts for

air temperature, humidity, solar radiation, and wind speed) of these laboratory conditions is often at or above 32.2°C. However, The Event Alert System (Road Runners Club of America) recommends road race cancellation in “hot” environmental conditions (WBGT > 32.2°C), and a recent analysis of medical events at the Twin Cities marathon suggested that running races should be canceled when starting weather is warm (WBGT > 22°C) to protect runner health and to avoid overwhelming local medical services (51). This means that runners, in real-world competitions, do not often compete in the environmental temperatures matching those of laboratory-based studies. However, there is strong evidence that a WBGT well below the range associated with race cancellation will still have a profound and detrimental impact on running performance (19,30,62).

Large temperature gradients between the body and the environment aid in the dissipation of metabolic heat. As the gradient between body temperature and the environment narrows, the ability to dissipate

heat is attenuated, more metabolic heat is stored, and body temperature therefore rises. The amount of heat storage that occurs during exercise is quantified using the human heat balance equation:

$$S = M - W \pm R \pm C - E,$$

where heat storage (S; gain in internal temperature) is the result of metabolic heat production (M) minus energy attributed to external work (W), with heat exchange through the environment quantified by radiation (R), convective heat (C) transfer, and evaporative heat loss (E; heat loss from sweat evaporation at the skin surface) (21). At the onset of exercise, a small amount of heat storage usually activates heat loss mechanisms including increased blood flow to the skin (cutaneous vasodilation) and stimulates the onset of thermoregulatory sweating. During moderate-intensity exercise in cool environments, these heat loss mechanisms are sufficient to maintain core temperature within a fairly narrow zone. However,

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because of the metabolic heat production of high-intensity running, it is not uncommon for core temperature to rise 2°C or more during half marathon or marathon competitions (5,9,38,52), even in relatively cool conditions. For example, the running pace of marathon world record (2:01:39 by Eliud Kipchoge) results in a metabolic heat production of ~1,560 W, which would require almost 5 L of sweat evaporation over the course of the marathon to maintain thermal equilibrium (assuming no heat is lost through R and C) (21). Given this substantial core temperature rise and evaporative requirement during running competition, it is unsurprising that performance is negatively impacted in environmental temperatures that would not generally be considered heat stress. Laboratory studies examining time to fatigue (22) noted a reduction in endurance at environmental temperatures as low as ~20°C, well below the conditions traditionally used in heat stress literature. While laboratory studies may not match real-world performance due to lower rates of convective cooling on a stationary cycle ergometer or treadmill as compared to over-ground running, a series of analyses in marathon performance further supports the notion that optimal performance conditions for distance running are comparatively cool.

A common weather parameter of fast running events, including marathons, is a relatively cool environmental air temperature of approximately ~12°C (8°C WBGT) (18,30,41,62). Small increases in air temperature beyond ~12°C lead to a progressive slowing of performance in elite, competitive, and casual runners with the slowest runners experiencing a greater reduction in performance (19,30,62). Outside of performance variables, the number of medical encounters including exertional heat-related illnesses has been shown to increase with small elevations in environmental conditions (>13°C WBGT, (30,51,62)). These data suggest that the high metabolic

rates of running in combination within warm environmental temperatures (12–22°C) can have a profound effect on physiology and running performance (Figure 1).

PHYSIOLOGY OF PERFORMANCE DECREMENTS IN HEAT

Degraded running performance and premature fatigue can be caused by a myriad of physiological and psychological factors. Investigators studying the performance-limiting physiology of heat stress historically focused on either the implications of an elevated brain temperature (7) or the profound cardiovascular demand of simultaneously perfusing active skeletal muscle and maintaining adequate skin blood flow for cooling (15). Some debate exists in exercise physiology on the primary reason that performance is diminished in warm-hot conditions.

The etiology of hyperthermia-induced fatigue seems to involve complex interactions between cardiovascular alterations, peripheral (muscular), and central factors (32). Contributing factors include elevated core temperature (28), elevated skin temperature and skin blood flow (26), convergence of core and skin temperature (47), progressive dehydration from sweat losses (54), and/or some combination of inputs to the brain to reduce heat production to avoid catastrophic blood pressure or internal temperature crises (anticipatory regulation) (37,61).

ELEVATED INTERNAL TEMPERATURE

The concept of diminished endurance or work capacity in the heat as a consequence of elevated core temperature was firmly established after several studies (28,42) demonstrated the

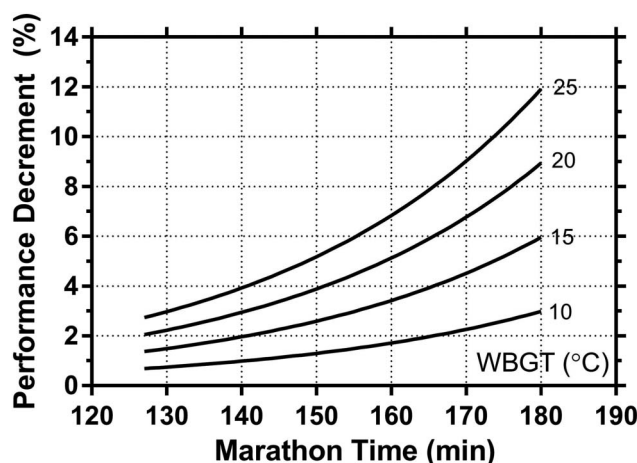


Figure 1. Nomogram presenting potential performance decrements (y axis) based on projected marathon finishing time (2:10–3:00 hours) with increasing WBGT temperature. A WBGT of 5–10°C represents the prevailing environmental conditions for the fastest performances for runners of all abilities. As the environmental conditions warm, performance slows in a quantifiable manner with the effects greater for the slower runners. For example, the qualifying standards for the 2020 Men’s and Women’s US Marathon trials are 2:19:00 and 2:45:00, respectively. Runners with the fitness to run these qualifying standards in near-ideal weather conditions (~5°C WBGT) would be expected to slow 1–2% for every 5°C increase in WBGT. At a WBGT of 15°C, this would equate to a 2% slowing in men (from 2:19:00 to 2:21:46) and a 4% slowing in women (from 2:45:00 to 2:51:36). The slowing is not sex-specific, as men and women slow similarly with increases warming environmental conditions and would therefore similarly apply to 2:19 female runners and 2:45 male runners. Reprinted with permission from Ely et al. 2007 (19).

reliable occurrence of heat stress fatigue at a “critical” core temperature of 40°C. This temperature is also associated with exertional heat stroke (2), as well as brain wave and motor neural output changes (43,46), consistent with an interpretation that a core temperature of ~40°C may be a safety brake to prevent catastrophic hyperthermia or at least mark the precipice of progressively plummeting performance (44,46). Although exercise performance is diminished with an elevated brain temperature ($\geq 40^\circ\text{C}$), the idea of a sharp performance precipice at a core temperature of 40°C is countered by numerous reports in the literature of athletes with rectal temperatures $>40^\circ\text{C}$ while apparently maintaining a high level of performance. This idea of a critical core temperature was investigated in highly fit runners performing an 8k track time trial where rectal temperature, skin temperature, and heart rate were continuously monitored in COOL (WBGT $\sim 13^\circ\text{C}$) and WARM (WBGT $\sim 27^\circ\text{C}$) conditions (17). Core temperature at completion of the 8k time trial was slightly higher in WARM (0.5°C higher than COOL) while skin temperature was substantially higher in WARM (4°C higher than COOL). Overall performance was substantially reduced in WARM conditions compared with COOL (runners in COOL averaged 1.5% slower than their personal best for 8k, while runners in WARM conditions averaged 7.9% slower than their personal best for 8k). However, mean running velocity when core temperature was under or over 40°C core did not differ for individuals in either WARM or COOL conditions. Furthermore, there was no evidence of runners adjusting pace based on the rate of core temperature rise in anticipatory avoidance of exceeding an internal temperature threshold.

Conversely, in an attempt to separate the influence of core and skin temperature on performance in the heat, a short-duration exercise performance task (15-minute time trial, similar to 3,000- to 5,000-m race performance)

was compared in hot (HOT; 40°C and 20% relative humidity) and temperate (TEMP; 20°C and 50% relative humidity) conditions where core temperature was matched between conditions and remained below 38.5°C, but mean skin temperature was markedly higher in hot conditions (+5°C; 31 versus 36°C). Importantly, this study also carefully controlled for hydration and found that both overall performance (Figure 2) and pacing strategy were substantially impaired in hot conditions (16). These studies suggest that core temperature may be overvalued as the primary factor in hot-weather performance decrements, while elevated skin temperature has a strong relationship with diminished performance. A corollary to the influence of elevated skin temperature is the idea that fatigue in the heat can occur with the convergence of core and skin temperature (8,10,47), thereby narrowing the temperature gradient that allows for heat exchange with the environment. The combination of core temperature and skin temperature inputs, as well as interindividual variability, results in a continuum of core temperatures (38–42°C) where heat-related performance impairment or exhaustion may occur (7).

ELEVATED SKIN TEMPERATURE AND SKIN BLOOD FLOW

During exercise in a cool environment, the autonomic nervous system constricts blood vessels to the skin to redistribute blood flow to active skeletal muscle while maintaining an adequate central volume (33). However, when an individual is running in a warm-hot environment, cutaneous vasoconstriction is withdrawn, and sympathetic vasodilation (active vasodilation) occurs, usually in concert with a sweating response (34). The vasodilation of the relatively compliant cutaneous circulation reduces central blood volume, stroke volume, and cardiac output (26). This is partially compensated for by a higher heart rate for a given running pace in warm-hot weather (1), but the increased heart rate is usually not adequate to perfuse

the skin to dissipate heat while maintaining blood flow to the working musculature without losses to blood pressure. In mild heat stress, the elevated heart rate needed to deliver blood to the skin in addition to the active musculature is related to both increased perception of effort and reduced performance (63). In more severe environmental heat stress, this redistribution of cardiac output and competition between skin and active skeletal muscle blood flow, coupled with the autonomic mechanisms in place to maintain central volume and blood pressure, results in a reduction in maximal oxygen delivery to muscle. This, in turn, leads to a reduction in maximal oxygen uptake ($\dot{V}\text{O}_{2\text{max}}$) in hot as compared to cool conditions (3,63).

DEHYDRATION

The competition for blood flow during running in the heat while simultaneously perfusing active skeletal muscle and warm skin described above is further exacerbated by fluid losses from thermoregulatory sweating. Loss of plasma volume coupled with competition for this progressively dwindling blood volume create a greater challenge to maintain venous return, central volume, cardiac output, and blood pressure over time. Runners competing in a warm ($\sim 25^\circ\text{C}$) half marathon race can easily lose 2–3% body mass through sweating, even while regularly consuming fluids at aid stations (48). In cool conditions, body water losses as low as 1–2% body mass can impair performance (55), and this performance decrement increases substantially as ambient temperature increases (35). While regular fluid intake during hot-weather running can minimize dehydration, ad libitum fluid intake often does not replace sweat losses. For example, trained females running 30k on a treadmill at marathon pace in warm, temperate, and cool environments replaced an average of $\sim 70\%$ of fluid losses, which resulted in $\sim 3\%$ body mass loss (6). A study examining higher fluid intakes during a warm, 2-hour run observed gastrointestinal discomfort in several athletes

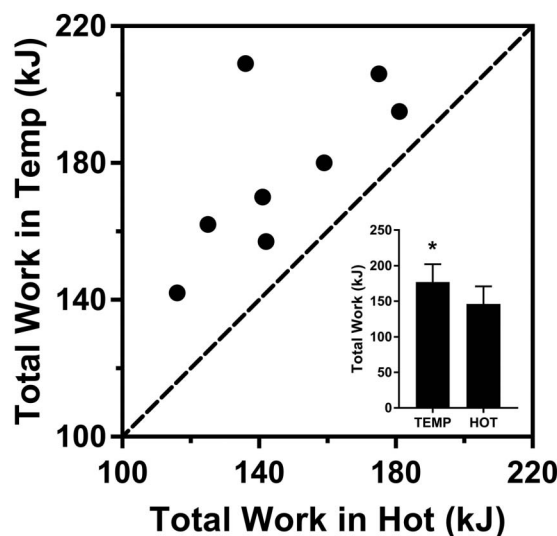


Figure 2. Total work completed during a 15-minute time trial in temperate (TEMP; 20°C) and hot (HOT; 40°C) conditions where core temperature did not differ between conditions and subjects were well-hydrated. Each dot represents a single individual in HOT (x axis) and TEMP (y axis), with the line of identity down the center indicating the point where performances are equivalent between HOT and TEMP. All subjects performed less work in HOT (all above line of identity) with an average impairment of 17% in HOT compared with TEMP (see figure inset). * denotes a significant ($p < 0.05$) difference between HOT and TEMP. This illustrates the substantial impact of heat stress on aerobic performance, even in the absence of elevated core temperature. Figure adapted with permission from Ely et al., 2010 (16).

(11), indicating that fully matching sweat losses with fluid replacement during running competition in the heat may be difficult. However, adequate precompetition hydration and in-competition ad libitum drinking can help minimize performance decrements associated with reduced central volume and hyperthermia in hot environments (60).

HEAT ACCLIMATION TO IMPROVE HOT-WEATHER RUNNING PERFORMANCE

Heat acclimation, or repeated exposure to exercise-heat stress, was developed in occupational settings to prolong the work capacity of individuals subjected to hot work conditions (31,53). These repeated exercise-heat exposures (60–100 minutes of moderate-intensity exercise in a warm/hot environment), over the course of 5–14 days, resulted in a reduced resting core temperature and slower rise during exercise, reduced heart rate at fixed workloads, increased sweating capacity, and increased plasma

volume (59). The improved ability to dissipate heat mitigates heat storage, and the slower rise in core temperature prolongs time to exhaustion during fixed-intensity exercise (42). Similarly, the improved ability to dissipate heat will allow for an elevated exercise intensity and metabolic heat production during self-paced exercise (49). In concert, these adaptations reduce thermoregulatory strain and improve physical performance in the heat.

Heat acclimation procedures developed in occupational settings translated easily to athletes preparing to compete in hot-weather races, and a variety of exercise-heat acclimation protocols and timelines seem to improve hot-weather running performance (49). Heat acclimation can be administered through use of an environmental chamber set to hot/dry or warm/humid conditions, through natural acclimatization by running outdoors in a warm/hot climate, or

potentially through the addition of insulative clothing during exercise in temperate conditions (overdressing) (14,15,58). Classic laboratory-based heat acclimation allows for precise control of conditions, while outdoor exercise in the heat may be more practical for teams preparing for hot-weather competition. Athletes who live and train in more temperate climates and do not have access to an environmental chamber may also attain some heat acclimation benefits through overdressing (12–15,58). However, consideration must be given to the clothing ensemble selected and the outdoor environment. Overdressing studies using heavier clothing and indoor treadmill running report high physiological strain (15) and enhanced heat tolerance (14), while studies using lighter clothing with outdoor exercise have reported smaller changes in heat gain (58) and no change in heat tolerance (57).

The process of heat acclimation requires repeated exposure to 3 critical stimuli: an elevated internal temperature (20), elevated skin temperature (50), and sweat production on the skin surface (4,27). Because athletes training at a high intensity in relatively cool environments likely experience all 3 of these stimuli on a regular basis, runners are believed to be “partially heat acclimated” as compared to sedentary counterparts (25). This means that a well-trained runner preparing for hot-weather competition can see thermoregulatory benefits of heat acclimation in as little as 5 days (23,24,39), while untrained individuals require a more extended (7–14 days) period of acclimation. Because exercise-heat exposure constitutes a greater training stress than running in cool conditions, the minimal exposure for benefit is an important consideration, particularly because heat acclimation is usually used before competition during a taper period (49).

The concern of overstressing a tapering athlete with the addition of exercise-heat acclimation has resulted in the innovation of passive heat acclimation

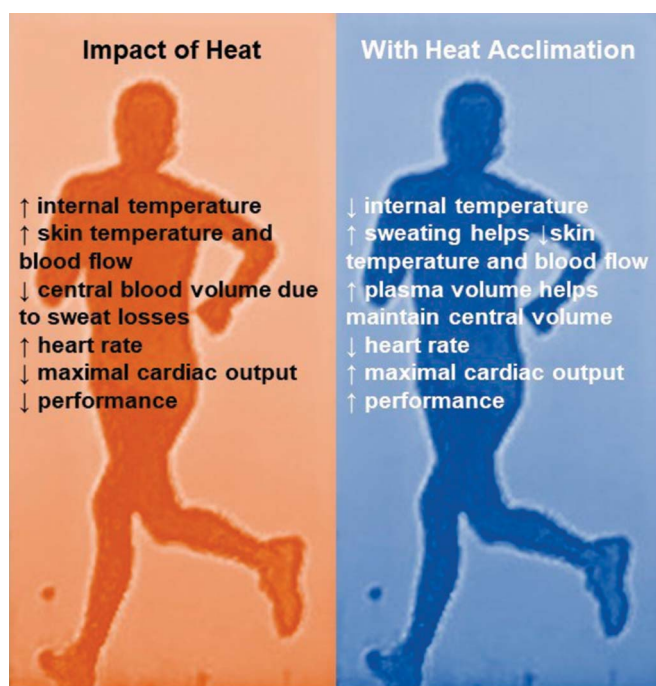


Figure 3. Summary of the physiological underpinnings of heat-related decrements in running performance, and the mechanisms through which heat acclimation can reduce thermoregulatory and cardiovascular strain to improve hot-weather running performance.

strategies (29). It is recommended that athletes incorporate passive heat acclimation bouts, including sauna or hot water immersion, immediately after exercise when core temperature is already elevated (29). Recent research has shown that as few as 6 consecutive days of hot water immersion after 40 minutes of treadmill running in temperate conditions reliably enhanced heat tolerance and expanded plasma volume in both highly fit (65) and modestly trained (64) male runners. Although performance in the heat was not specifically investigated, 3 weeks of post-exercise humid sauna exposure in trained male runners increased plasma volume and enhanced performance in a simulated 5,000-m run in temperate conditions (56). The increased plasma volume, in particular, may contribute to improved maximal performance in cool conditions as well (36,56), although this is not universally noted (40,45).

Thus, the physiological underpinnings of performance decrements in the heat can all be mitigated by heat

acclimation. The reduced basal temperature and slower rise during exercise reduces hyperthermia, the increased sweating response facilitates heat exchange through a widened core to skin temperature gradient, and the increased plasma volume enables a greater skin and muscle blood flow and increases the sweating response (Figure 3). In mild heat stress, this means that performance can be likely maintained as compared to cooler conditions with heat acclimation. However, with more severe heat stress, heat acclimation mitigates but does not abolish heat-associated performance decrements (36).

SUMMARY

The profound metabolic heat production of high-intensity running must be countered by adequate heat loss to the environment. As the gradient between body temperature and the environment narrows, the ability to dissipate heat is attenuated

and more metabolic heat is stored, and body temperature rises. This means that the environment can begin to impact running performance at relatively low-ambient temperatures, and running performance continues to slow as ambient temperature increases. The primary causative factors involve elevated internal temperature, high skin blood flow, and progressive fluid losses through thermoregulatory sweating. Heat acclimation, through repeated active or passive exposures, can help minimize the performance decrements associated with the heat.

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